Exp. No.: 04
DATE: 21/9/25
Pg. No.: 18

DESIGN A REGULATED POWER SUPPLY CIRCUIT USING A VOLTAGE REGULATOR IC

AIM:

To design a regulated power supply circuit using a voltage regulator IC, simulate the circuit in LTSpice, and test its performance for different load conditions.

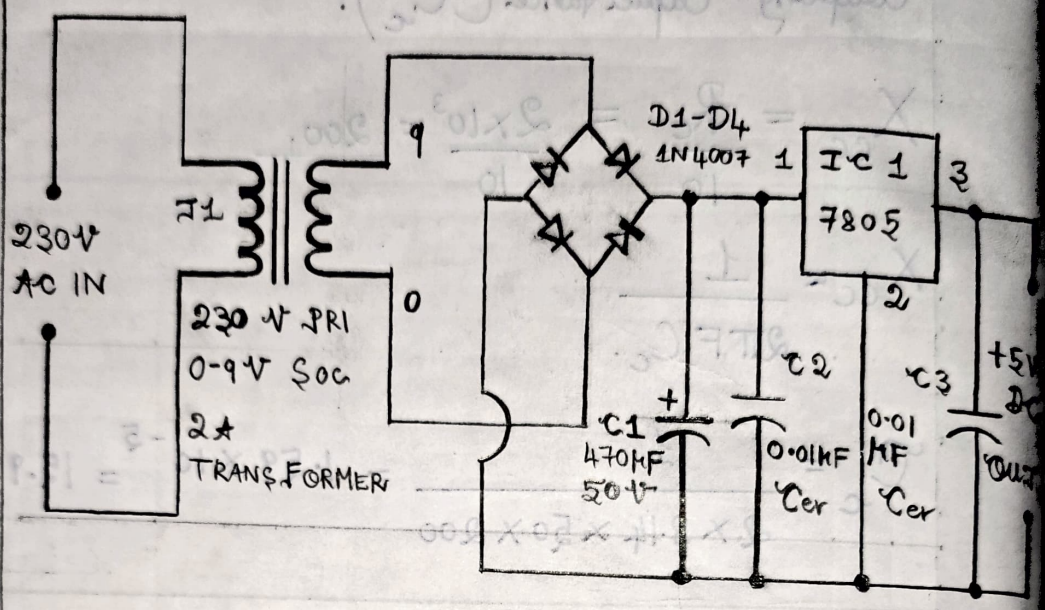
SOFTWARE REQUIRED:

- LTSpice Simulator.

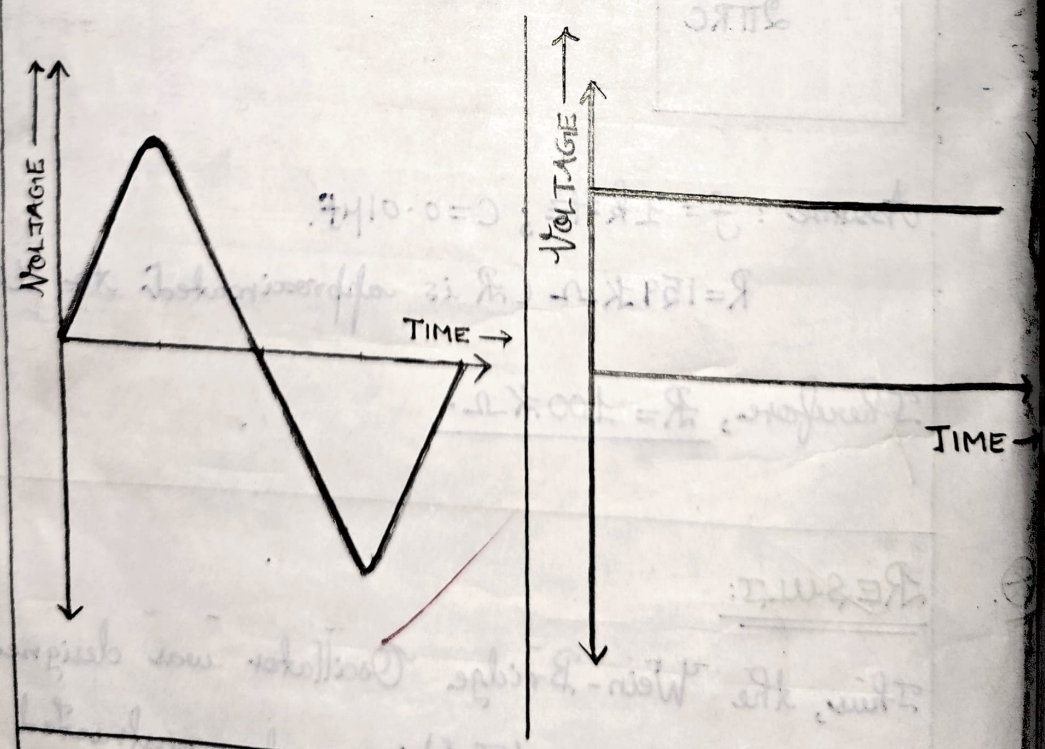
PROCEDURE:

1. Connecting the circuit on bread board as per the circuit diagram.
2. Connect the primary of the transformer to main supply, i.e, 230V, 50Hz.
3. Connect the elements and complete the circuit.
4. Measure the output by varying the input load of various values.
5. Compares the output for different load values.

CIRCUIT DIAGRAM:



MODEL GRAPH:



⊛. TABULATION:

SI.NO.	INPUT VOLTAGE (V_{IN})	INPUT FREQUENCY (Hz)	OUTPUT VOLTAGE (V_o)
1.	15.2	50	5.01
2.	15.5	50	5.00
3.	16.0	50	5.02
4.	16.3	50	5.01
5.	16.8	50	5.00
6.	17.2	50	5.02
7.	17.5	50	5.01
8.	17.8	50	5.00
9.	18.0	50	5.01
10.	18.2	50	5.00

⊛. CALCULATION:

i). LINE REGULATION (L.R):

Eq: $L.R = \frac{\Delta V_{out}}{\Delta V_{in}}$

→ Eq: $L.R = \frac{5.02 - 5.00}{18.2 - 15.2} = \frac{0.02}{3} = 0.0067.$

RESULT:

Thus, a regulated power supply circuit using a voltage regulator IC was designed and the circuit was simulated using LT Spice and its performance for different load conditions was tested.